

Software Installation and Management





SOFTWARE INSTALLATION AND MANAGEMENT

SMP/E

System Modification Program Extended

The package manager for z/OS

z/OSMF

z/OS Management Facility

Browser-based front-end to z/OS system management tools; also exposes APIs used by external tools.

Ansible

An external tool that can interact with z/OSMF via ssh.



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We are here



WHAT TYPE OF SOFTWARE IS SMP/E?



SMP/E is a *package manager*.

Installing software on a computer system comes down to placing certain files on peripheral devices where the computer system can find them, and preparing executables to be loaded and executed on the system.

It's possible to install software on any computer system by manually copying files and building executables. Any system that's used in a serious way will quickly become an unmanageable mess if people try to manage the system in that way, whether it's someone's personal device or a large-scale corporate system.

Quite early in the history of the software industry, people developed software that installs other software. It keeps track of dependencies between software products and manages the sequence of installation or removal of software products. They're called *package managers*.

You've worked with more than one package manager in your career so far, and probably in your personal life, too.

SMP/E is the one used for z/OS systems.

Other IBM operating systems that can run on the IBM Z have their own package managers.

WHAT'S A PACKAGE?

So, we're managing packages. Great. What's a *package*, then?

Some computer programs might be self-contained in a single file. They don't need any libraries or other products installed on the system. But that's the exception, not the norm.

Most software products comprise multiple files of various kinds – source code, object code, executables, documentation, configuration files, internal databases, GUI templates, localization values, examples to help users learn to work with the product, and more.

The collection of files necessary to make a software product usable is called a *package*. A package manager keeps track of versions, dependencies, and other details about packages.

The good news is IBM uses the same buzzword for this as the rest of the world: *package*. The bad news is there are more buzzwords coming soon.

Here's one now: *element*. The individual items contained in a package are called *elements*.



SYSTEM MODIFICATIONS



When you change any of the software on a computer system in any way (new install, upgrade, bug fix, uninstall), you're modifying the system.

Up until roughly 2000, IBM tended to name their products by the job they did. The original version of z/OS was called "Operating System." IMS stands for "Information Management System." DB2 means "Data Base," too.

They didn't name their products things like "Python" or "Homebrew" or "Ansible." You don't have to guess that System Modification Program is for modifying the system.

And the modifications to the system are called *system modifications*, or SYSMODs.

CATEGORIES OF SYSMODS

There are four types of SYSMODs.

- **Function SYSMOD** – install a new product or a new version or release of a product that's already present on the system.
- **PTF SYSMOD** – Program Temporary Fix or Product Temporary Fix (PTF) is an IBM-supplied correction for a known bug.
- **APAR SYSMOD** – Another kind of fix. IBM calls a bug report an APAR (Authorized Program Analysis Report). This type of SYSMOD fixes an APAR. APARs can be pretty minor, while PTFs are more significant.
- **USERMOD** – A "user" modification to a product (IBM or other), such as a user exit routine. These need to be tracked and managed because user modifications create differences between the base product and the product as customized in a particular installation. Subsequent SYSMODs for the base product shouldn't destroy the user's customization. Also, a user modification can introduce a dependency that the base product alone doesn't have. That has to be managed.



FUNCTION MODIFICATION ID (FMID)

SYSMODs are uniquely identified by a value called the Function Modification ID, or FMID (pronounced "eff-mid").

This information is dynamic and changes frequently. The z/OSMF facility, which front-ends SMP/E, has a screen that lets you query the FMIDs that are currently on your own system.

The illustration at right came from an obsolete list of FMIDs that IBM used to maintain as a public reference. They found it was tedious to keep the list up to date, and not that useful to customers after all. It gives you an idea of what FMIDs look like and how they relate to products.

FMID	NAME				
EAS1102	ASSEMBLER XF	5752SC103	R102	MVSS038J	
EB81102	BASE CONTROL PROGRAM	5752SC***	R102	MVSS038J	
EBT1102	BTAM	5752SC120	R102	MVSS038J	
EDE1102	DEMF	5752CM100	R102	MVSS038J	
EDM1102	DATA MANAGEMENT	5752SCID*	R102	MVSS038J	
EDS1102	DASD ERP	5752SC1C*	R102	MVSS038J	
EDU1602	ICKDSF R068 MVS/XA	565525701	R602	ICKDSF060	
EDU1702	ICKDSF R070 MVS/XA	565525701	R702	ICKDSF070	
EDU1802	ICKDSF R080 MVS/XA	565525701	R802	ICKDSF080	
EEP0100	370X-EP PEP	5735SC100	R100	EP030	
EEP0101	EP/OS	5735SC100	R101	EP030	
EEP0200	EP/OS WITH ACF/NCP R2	5735SC100	R200	EP030	
EEP0300	EP FOR NCP REL3	5735SC100	R300	EP030	

MAKING SYSMODS AND APPLYING THEM

There are two things people do with SMP/E.

System programmers who work at companies that are IBM customers use it to install and update software packages from IBM and independent software vendors (ISVs). This may be part of the work of a Platform Engineer at your company.

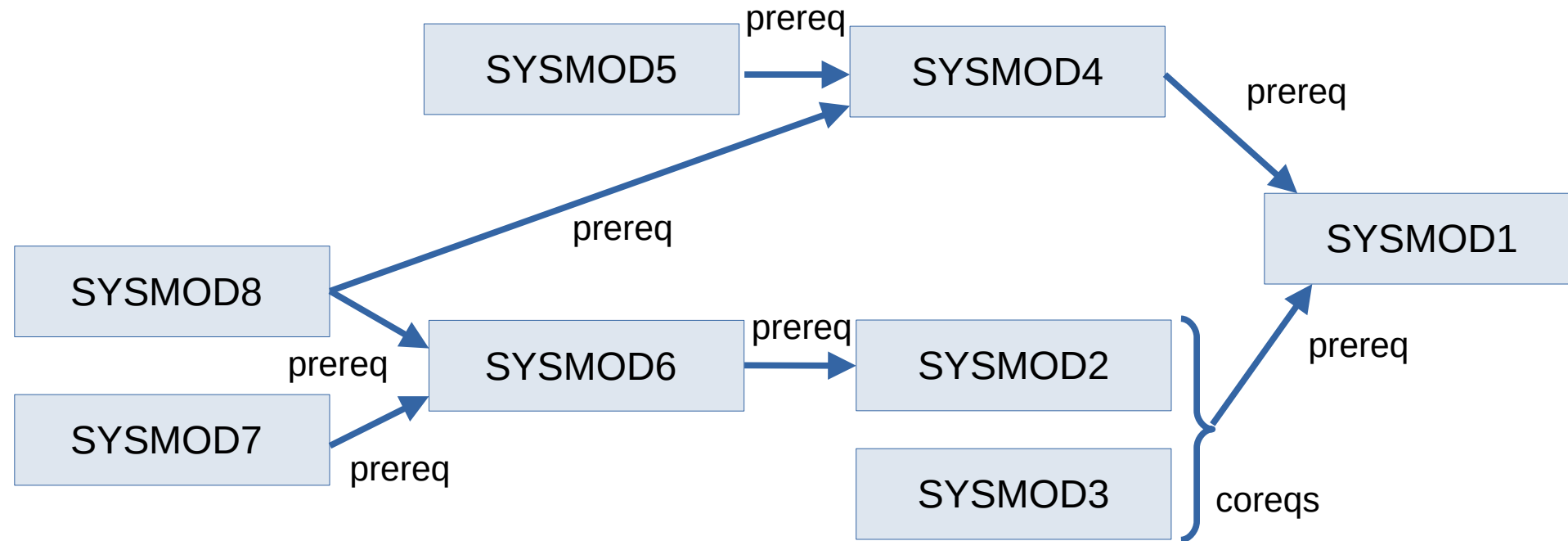


IBM and ISV's use SMP/E to prepare packages according to IBM's standards so that customers can install their products using SMP/E. As a Platform Engineer at a company that uses IBM products, you will not have to do this. This is done by IBM and independent software vendors (ISVs).



REQUISITE CHAINS

PREREQ (prerequisite) must be installed before a given SYSMOD
COREQs (corequisites) must be installed together



The illustration suggests our goal is to install SYSMOD1. To do so, we must ensure the prerequisite SYSMODs are installed in the correct order, and corequisites are installed together. The whole sequence is called the *requisite chain* for SYSMOD1.

SMP/E MODIFICATION CONTROL STATEMENTS

SMP/E runs as a batch program that is controlled by control statements called *Modification Control Statements* (MCS). The MCS function is a declarative language we use to direct SMP/E to perform its various functions.

Define an existing PDS as a TLIB.

```
//SMPCNTL DD *  
  SET BDY(TSTTGT) .  
  
  UCLIN .  
    ADD DDDEF(TGTLOAD)  
      DA('HLQ.TSTTGT.LOAD') .  
  ENDUCL .  
/*
```

See what would happen if changes were applied.

```
//SMPCNTL DD *  
  SET BDY(TSTTGT) .  
  APPLY CHECK  
    S(UM00001) .  
/*
```

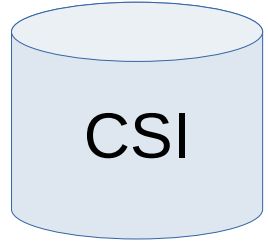
Receive SYSMODs.

```
//SMPCNTL DD *  
  SET BDY(GLOBAL) .  
  RECEIVE SYSMODS .  
/*
```

Apply the changes.

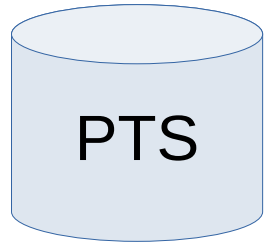
```
//SMPCNTL DD *  
  SET BDY(TSTTGT) .  
  ACCEPT CHECK  
    S(UM00001) .  
/*
```


SMP/E DATA SETS



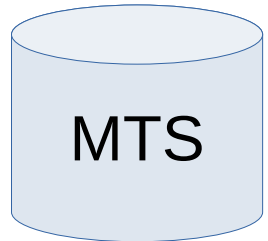
Consolidated Software Inventory

SMP/E uses the CSI to track the distribution and target libraries.



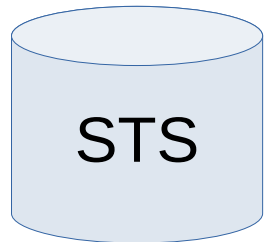
PTF Temporary Storage

This data set holds PTF SYSMODs as received.



Macro Temporary Storage

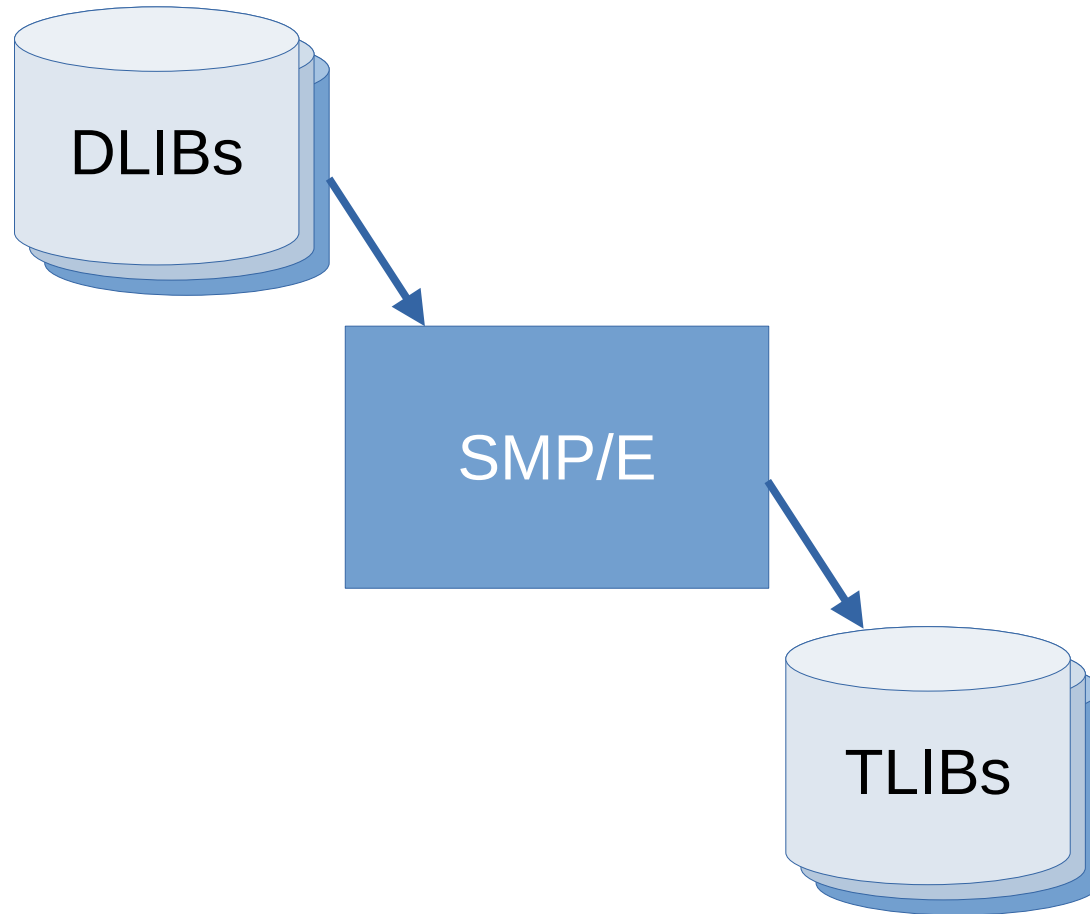
This data set holds macros used in assemblies during APPLY processing.



Source Temporary Storage

This data set holds source code used in assemblies during APPLY processing.

DLIBS AND TLIBS



SMP/E works with two types of libraries: *Distribution Libraries* (DLIBs) and *Target Libraries* (TLIBs). They aren't physically different types of artifacts; the distinction is in how they're used.

A DLIB contains the master copy of the elements constituting each product on a system.

A TLIB is a target of the product installation process. It contains executables and other types of files that are needed to operate the product on the system.

DISTRIBUTING SOURCE



As you read the IBM documentation for SMP/E, you'll notice there's an unstated assumption that the SYSMODs were prepared on another, existing z/OS system.

Many are object code only (OCO) distributions, which means their sources were assembled or compiled into object modules, and the object modules were bound or link-edited into executables or load modules, before the SYSMOD was published.

That will be the case for components of z/OS itself, and many of the utilities that ship with it.

That can only happen on another z/OS system. There's no way to create z/OS executables on other operating systems.

You can also distribute a product in source form and specify that the code has to be compiled or assembled as part of the SMP/E installation process.

BASIC INSTALLATION STEPS

Although the underlying details are complicated, the steps to follow when installing a SYSMOD are straightforward.

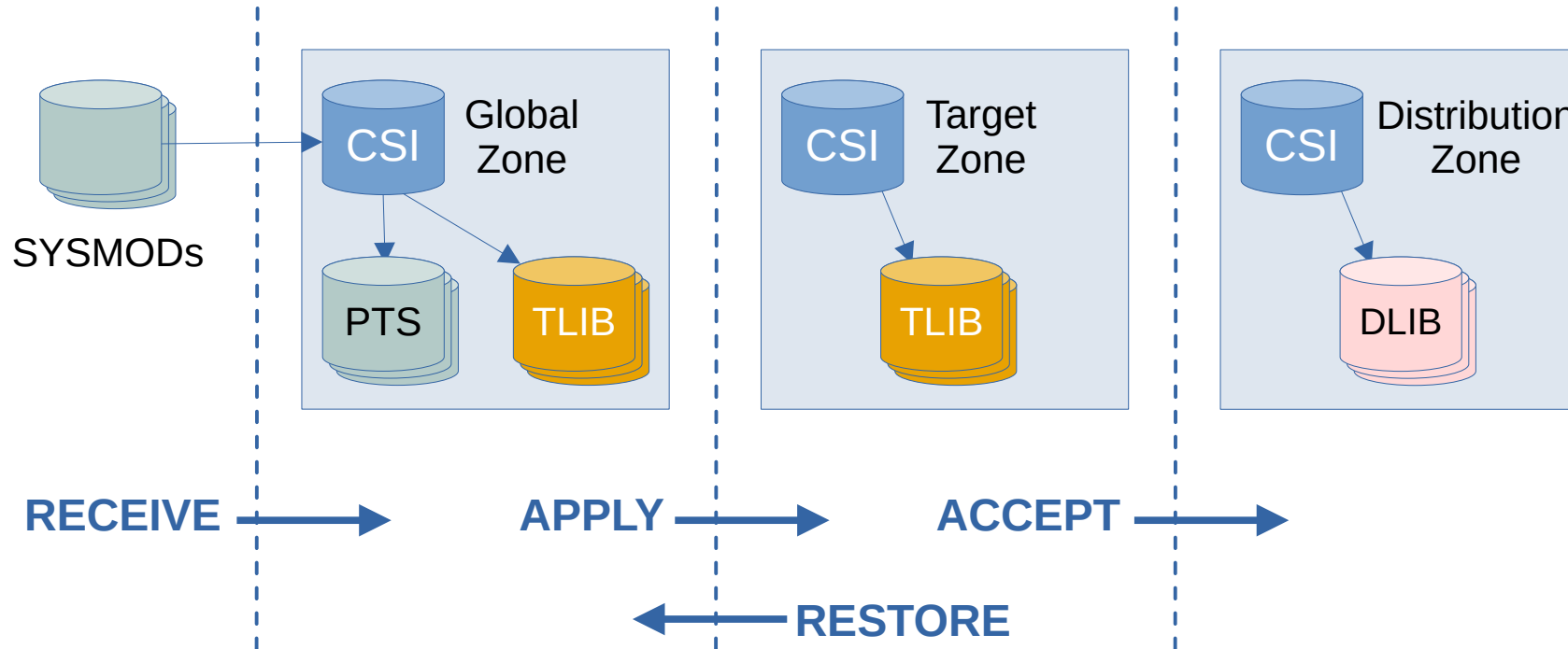


Illustration loosely based on IBM documentation: <https://www.ibm.com/docs/en/zos-basic-skills?topic=smpe-process-installing-zos-elements-service>

INITIAL SMP/E ISPF DISPLAY

```

----- SMP/E PRIMARY OPTION MENU ----- SMP/E 37.22
=> █

0 SETTINGS      - Configure settings for the SMP/E dialogs
1 ADMINISTRATION - Administer the SMPCSI contents
2 SYSMOD MANAGEMENT - Receive SYSMODs and HOLDDATA
                    and install SYSMODs
3 QUERY         - Display SMPCSI information
4 COMMAND GENERATION - Generate SMP/E commands
5 RECEIVE       - Receive SYSMODs, HOLDDATA and
                    support information
6 ORDER MANAGEMENT - Manage ORDER entries in the global zone

D DESCRIBE      - An overview of the dialogs
T TUTORIAL      - Details on using the dialogs
W WHAT IS NEW   - What is New in SMP/E

Specify the name of the CSI that contains the global zone:
SMPCSI DATA SET  =>
(Leave blank for a list of SMPCSI data set names.)

Specify YES to have DD statements for SYSOUT and temporary
data sets generated. Specify NO, to use DDDEFs.
Generate DD statements  => NO

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```

The ISPF panels offer a good way to begin to get familiar with SMP/E. In the upper right-hand corner, you see the version of SMP/E that's installed on this z/OS instance.

When you're looking for documentation, online resources, or help from humans or LLMs, you need to know the SMP/E version. Commands and behavior vary between versions, sometimes significantly.

You can choose from among several options. Two of them will be useful to get acquainted with the facility: D for Describe and T for Tutorial.

We'll leave it to you to check those out during the next lab exercise.

For now, let's try a couple of the options and see what they do.

FINDING CSI DATA SETS

```
----- SMP/E PRIMARY OPTION MENU ----- SMP/E 37.22
=> █

0 SETTINGS      - Configure settings for the SMP/E dialogs
1 ADMINISTRATION - Administer the SMPCSI contents
2 SYSMOD MANAGEMENT - Receive SYSMODs and HOLDDATA
                  and install SYSMODs
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(Leave blank for a list of SMPCSI data set names.)

Specify YES to have DD statements for SYSOUT and temporary
data sets generated. Specify NO, to use DDDEFS.
Generate DD statements => NO

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```

If you aren't sure where to start or you don't know the names of CSI data sets, you can choose option 3, Query, and leave the name of the SMPCSI data set blank.

FINDING CSI DATA SETS

That gets you here, to a list of the CSI data sets SMP/E knows about on this z/OS instance.

SMPE.ZOS31.CSI looks promising. The name suggests it's fundamental to the system.

```
----- SMP/E CSI SELECTION -----  
=>   
  
Modify the CSI data set name list? NO_ (Yes/No)  
  
Select the name of the CSI to be processed.  
  
Primary command: FIND  
Line command: S - Select  
  
S          SMPCSI Data Set Name  
-  -----  
-  'MQCD93.CSI'  
-  'SMPE.PROGPROD.CSI'  
-  'SMPE.ZOS31.CSI'  
-  'SMPEDEV.PROGPROD.CSI'  
***** Bottom of data *****
```



QUERYING CSI DATA

Next, we're presented with a few options to perform queries on the CSI data. Let's try option 1, CSI Query.

```
==> CSI QUERY

Specify the zone, entry type, and name to be queried:

ZONE NAME ==> global Name of the zone to be queried.
To display a list of all zones,
leave blank

ENTRY TYPE ==> Entry type to be queried.
To display a list of all valid
entry types, leave ENTRY TYPE
and ENTRY NAME blank

ENTRY NAME ==> Entry name to be queried.
Leave blank or use a wildcard
(entry name pattern) to display
a selection list.

To return to the Query selection menu, enter END .
```

```
==> 1 QUERY SELECTION MENU

1 CSI QUERY - Display SMPCSI entries
2 CROSS-ZONE QUERY - Display status of an entry in
all zones
3 SOURCEID QUERY - Display SOURCEIDs for specified zone
D DESCRIBE - Overview of using QUERY
T TUTORIAL - Information on using QUERY

To return to the SMP/E primary option menu, enter END .
```

We don't know enough yet to specify exactly what we're looking for. Let's enter "global" zone and leave the other options blank.

QUERYING DDDEF DATA

```

      CSI QUERY - ENTRY TYPE SELECTION
      ==>
      Select one entry type for GLOBAL zone GLOBAL
      To display a list of all entries, leave the NAME field blank

      S  TYPE      NAME      ACTION
      s  DDDEF
      FEATURE
      FMIDSET
      GZONE
      HOLDDATA
      MCS
      OPTIONS
      ORDER
      PRODUCT
      SYSMOD
      UTILITY
      ZONESET
      ***** Bottom of data *****

```

Let's try each option and see what kinds of information each one provides.

The first one listed is DDDEF. We don't know the names of any of them, so we'll leave that field blank and see what we get.

QUERYING DDDEF DATA

```
CSI QUERY - SELECT ENTRY          Row 1 to 19 of 19
SCROLL ==> PAGE

Select one entry to query from GLOBAL zone GLOBAL :

S  NAME      ACTION
  ASMPRINT
  BPXPRINT
  SMPDEBUG
  SMPJHOME
  SMPLIST
  SMPLOG
  SMPLOGA
  SMPOUT
  SMPPTS
  SMPPUNCH
  SMPRPT
  SMPSNAP
  SMPTLIB
  SYSPRINT
  SYSPUNCH
  SYSUT1
  SYSUT2
  SYSUT3
  SYSUT4

***** Bottom of data *****
```

So we get a list of DDDEFs from the GLOBAL zone in data set SMPE.ZOS31.CSI.
Let's see what ASMPRINT look like.

QUERYING DDDEF DATA

There's not much to see here. It might be here as a placeholder or example.

We're just exploring, to learn what we can about SMP/E. Even this empty form tells us something – the kinds of information a DDDEF stores.

But you don't need an instructor to walk you through a process of exploration. You can do that in the next lab.

```

                                CSI QUERY - DDDEF ENTRY                                Row 1 to 1 of 1
=> █                                                                    SCROLL => PAGE

To return to the previous panel, enter END .

Primary Command: FIND

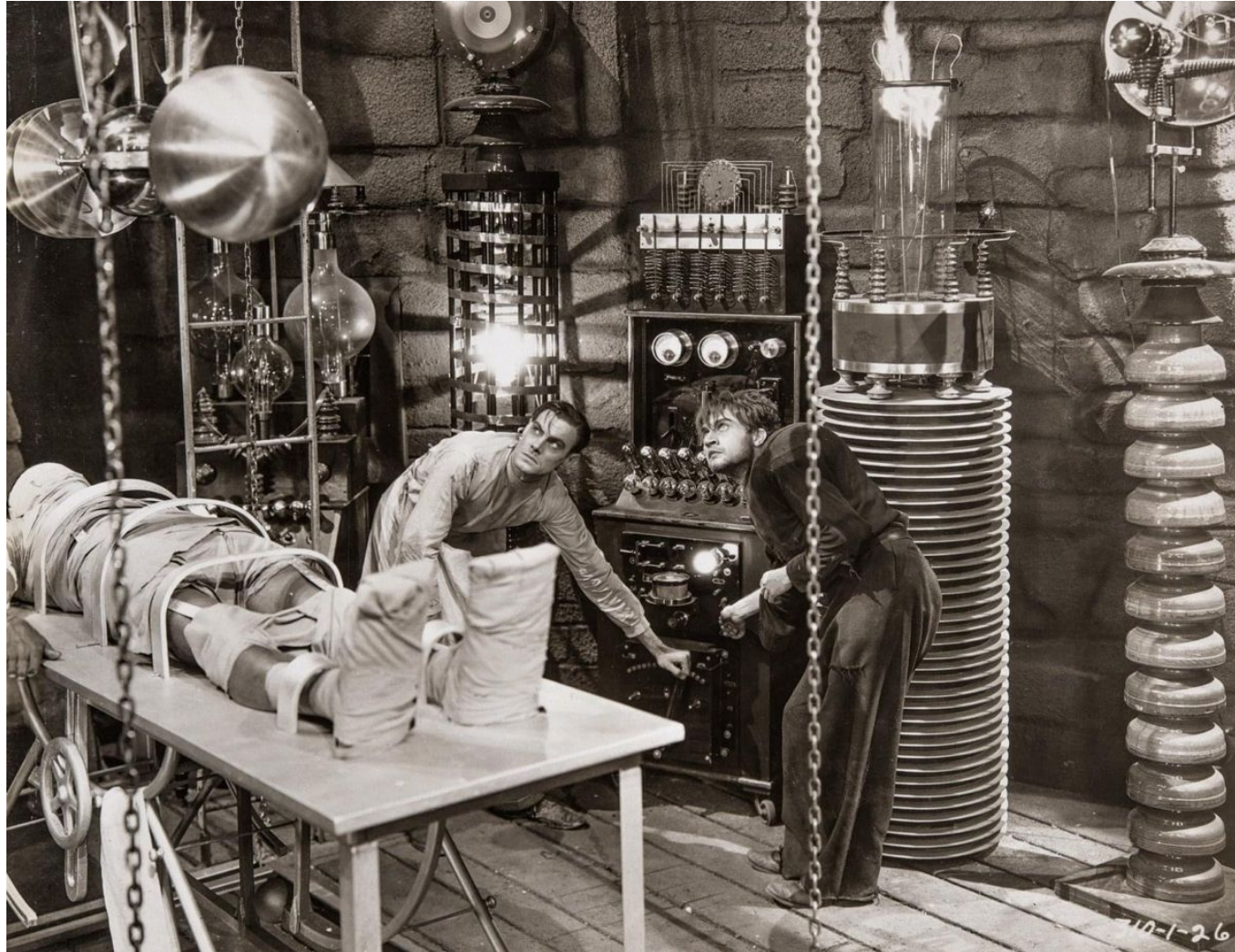
Entry Type:  DDDEF                                Zone Name: GLOBAL
Entry Name:  ASMPRINT                             Zone Type:  GLOBAL


VOLUME:      UNIT:      DISP:      DIR:
SPACE :      PRIM:      SECOND:
SYSOUT CLASS: *      WAITFORDSN:
PROTECT:
DATACLAS:      MGMTCLAS  :
STORCLAS:      DSNTYPE   :
-----

***** Bottom of data *****

```

LAB Z/OS 37: EXPLORE SMP/E ISPF MENUS



A CLOSER LOOK AT THE CSI

A Consolidated Software Inventory (CSI) data set controls SMP/E processing and tracks the status and results of installation activities.

The information is divided into three categories called *zones*:

Global Zone – there's exactly one per CSI. This holds information about the SYSMODs.

DLIB Zone – there's at least one per CSI. This zone tracks the Distribution Libraries (used when building the artifacts to be installed).

Target Zone – there's at least one per CSI. This zone tracks the target libraries (where the installed product lives).



Global zone shown near upper right-hand corner of photo.

A CLOSER LOOK AT THE CSI – GLOBAL ZONE

1

QUERY - ZONE SELECTION

Row 1 to 3 of 3

⇒

Select the zone that you want to query:

S	NAME	TYPE	CSI DATA SET
	DLB31	DLIB	SMPE.ZOS31.CSI
s	GLOBAL	GLOBAL	SMPE.ZOS31.CSI
	TGT31	TARGET	SMPE.ZOS31.CSI

***** Bottom of data *****

2

Select one entry type for GLOBAL zone
To display a list of all entries, leave blank

S	TYPE	NAME	ACTION
	DDDEF		
s	FEATURE		
	FMIDSET		

3

Select one entry to query

S	NAME	ACTION
	DZ0BB310	
s	DZ0BJ310	
	IZ0BB310	

4

CSI QUERY - FEATURE ENTRY

Row 1 to 4 of 4

⇒

SCROLL ⇒ PAGE

To return to the previous panel, enter END .

Primary Command: FIND

Entry Type: FEATURE Zone Type: GLOBAL

Entry Name: DZ0BJ310

Description: z/OS V3 Base JPN OPT. Features Installed with Base

Product: 5655-ZOS 03.01.00

Date/Time: 24.017 17:49:07 REC

FMID	HCS77E0	HGD3201	HLB77C0	HMOS705	HNET7D0	HOP17D0	HQX77E0
	HRF77E0	HRM77E0	HSM1310	H24P111	JLB77CJ	JMOS7J5	JMQ416A
	JNET7DJ	JOP17DJ	JRF77EJ	JRM77EJ	J24P112		

REWORK 2023025

***** Bottom of data *****

Here you can see three zones exist in SMPE.ZOS31.CSI. Within each zone, information about one or more entry types is maintained. Zero or more entries are stored under each entry type.

When querying in ISPF, you first choose the CSI data set, then the zone within that data set, then the entry type within that zone, and finally the entry you want to examine.

A CLOSER LOOK AT THE CSI – DLIB ZONE

1 QUERY - ZONE SELECTION Row 1 to 3 of 3
SCROLL ==> CSR

Select the zone that you want to query:

S	NAME	TYPE	CSI DATA SET
S	DLB31	DLIB	SMPE.ZOS31.CSI
	GLOBAL		
	TGT31		

2 Select one entry type
To display a list of

S	TYPE	NAME
S	DLIB	
S	DZONE	
	EXEC	

3 CSI QUERY - SELECT ENTRY Row 36 to 41 of 41
SCROLL ==> CSR

Select one entry to query from DLIB zone DLB31 :

S	NAME
S	AMODGEN
S	ANUCLEUS
	APARMLIB
	APROCLIB
	ASAMPLIB
	ATSOMAC

4 CSI QUERY - DLIB ENTRY Row 1 to 1 of 1
SCROLL ==> CSR

To return to the previous panel, enter END .

Primary Command: FIND

Entry Type: DLIB Zone Name: DLB31
Entry Name: ANUCLEUS Zone Type: DLIB

LASTUPD: HBB77E0 TYPE=ADD

Now let's back up to the Zone Selection panel and choose the DLIB Zone (Distribution Zone). Then we choose Entry Type DLIB, and we'll look at the ANUCLEUS entry. This is obviously not the actual Nucleus code. The CSI contains labels or names of things that cross-reference entries in other Zones. SMP/E uses this information to guide its processing.

A CLOSER LOOK AT THE CSI – DLIB ZONE

The image shows three overlapping screenshots of the CSI (Control System Interface) system, illustrating the steps to view detailed information for a specific entry in the DLIB zone.

Panel 1 (Left): QUERY - ZONE SELECTION
 Row 1 to 3 of 3
 SCROLL ==> CSR
 Select the zone that you want to query:

S	NAME	TYPE	CSI DATA SET
S	DLB31	DLIB	
	GLOBAL	GLOBA	
	TGT31	TARGE	

Panel 2 (Middle): CSI QUERY - ENTRY TYPE SELECTION
 Row 152 to 185 of 746
 Select one entry type for DLIB
 To display a list of all entries:

S	TYPE	NAME	ACT
S	DLIB		
s	DZONE		
	EXEC		

Panel 3 (Right): CSI QUERY - ZONE ENTRY
 Row 1 to 1 of 1
 SCROLL ==> CSR
 To return to the previous panel, enter END .
 Primary Command: FIND
 Entry Type: DZONE
 Entry Name: DLB31
 Zone Name: DLB31
 Zone Type: DLIB
 Default OPTIONS: ZOSOPT
 UPGRADE LEVEL: SMP/E 37.16
Related Zone: TGT31
 JCLIN at ACCEPT: YES

 SRELS Z038

Still in the DLIB Zone, we select Entry Type DZONE and look at entry DLB31. Like the previous example, this doesn't show detailed information about the distribution elements. We can see a reference to a Target Zone, TGT31. That's the name of a Target Zone in the same CSI (you can see it listed on the Zone Selection panel screenshot above). There can be more than one DLIB Zone and Target Zone in a CSI, so SMP/E needs to know which one goes with this entry.

A CLOSER LOOK AT THE CSI – TARGET ZONE

1

QUERY - ZONE SELECTION

Row 1 to 3 of 3

SCROLL ==> CSR

Select the zone that you want to query:

S	NAME	TYPE
	DLB31	DLIB
	GLOBAL	GLOBAL
s	TGT31	TARGET

2

CST DATA SET

CSI QUERY - ENTRY TYPE SELECTION

Row 151 to 184 of 746

SCROLL ==> CSR

Select one entry type for TA

To display a list of all ent

S	TYPE	NAME	ACT
	DDDEF		
s	DLIB		
	EXEC		

3

CSI QUERY - DLIB ENTRY

Row 1 to 1 of 1

SCROLL ==> CSR

To return to the previous panel, enter END .

Primary Command: FIND

Entry Type: DLIB

Entry Name: ANUCLEUS

Zone Name: TGT31

Zone Type: TARGET

LASTUPD: HBB77E0 TYPE=ADD

SYSLIB NUCLEUS

***** Bottom of data *****

Looking at Target Zone TGT31 in the same CSI, we choose Entry Type DLIB and entry ANUCLEUS to see what kind of information the CSI contains in the Target Zone.

DATA SETS ASSOCIATED WITH SMPE.ZOS31

```

Data Set List Utility

Option ==> █

blank Display data set list      P Print data set list
  V Display VTOC information      PV Print VTOC information

Enter one or both of the parameters below:
Dsname Level . . . SMPE.ZOS31
Volume serial . . .
  
```

```

DSLIST - Data Sets Matching SMPE.ZOS31      Row 1 of 11
Command ==> █      Scroll ==> PAGE

Command - Enter "/" to select action      Message      Volume
-----
SMPE.ZOS31.CSI                          *VSAM*
SMPE.ZOS31.HOLDDATA                     T31VS1
SMPE.ZOS31.IDX.INDEX                    T31VS1
SMPE.ZOS31.KSD.DATA                     T31VS1
SMPE.ZOS31.SMPLOG                       T31VS1
SMPE.ZOS31.SMPLOGA                      T31VS1
SMPE.ZOS31.SMPLTS                       T31VS1
SMPE.ZOS31.SMPMTS                       T31VS1
SMPE.ZOS31.SMPPTS                       D31VS1
SMPE.ZOS31.SMPSCDS                      T31VS1
SMPE.ZOS31.SMPSTS                       T31VS1
***** End of Data Set list *****
  
```

SMP/E uses several other data sets besides the CSI to manage system modifications. Here we can see all the data sets that existed on the lab system as of the time the screenshots were taken whose HLQ and MLQ match those of the CSI data set, SMPE.ZOS31.CSI.

SMP/E creates these data sets when SYSMODs are received and processed, but all of them may not be used in every case.

DATA SETS ASSOCIATED WITH SMPE.ZOS31

```

Data Set List Utility

Option ==> █

blank Display data set list      P Print data set list
  V Display VTOC information      PV Print VTOC information

Enter one or both of the parameters below:
Dsname Level . . . SMPE.ZOS31
Volume serial . . .

```

```

DSLIST - Data Sets Matching SMPE.ZOS31      Row 1 of 11
Command ==> █      Scroll ==> PAGE

Command - Enter "/" to select action      Message      Volume
-----
SMPE.ZOS31.CSI                          *VSAM*
SMPE.ZOS31.HOLDDATA                     T31VS1
SMPE.ZOS31.IDX.INDEX                    T31VS1
SMPE.ZOS31.KSD.DATA                     T31VS1
SMPE.ZOS31.SMPLOG                       T31VS1
SMPE.ZOS31.SMPLOGA                      T31VS1
SMPE.ZOS31.SMPLTS                       T31VS1
SMPE.ZOS31.SMPMTS                       T31VS1
SMPE.ZOS31.SMPPTS                       D31VS1
SMPE.ZOS31.SMPSCDS                      T31VS1
SMPE.ZOS31.SMPSTS                       T31VS1
***** End of Data Set list *****

```

When we say a SYSMOD has been *received*, we don't mean that it has been stored in a data set on our system. We mean the SMP/E RECEIVE process has been done.

At that time, SMP/E creates CSI records for it, creates the SMPPTS data set, and stores the SYSMOD in that data set in the form of *MCS Members*.

The data set has a special format that is only used by SMP/E. You can browse its contents, but it will look "funny." There may be sections of plain text and sections of object code.

It is a kind of "library" in the general sense, but it is not a PDS or PDS/E.

DATA SETS ASSOCIATED WITH SMPE.ZOS31

```

Data Set List Utility

Option ==> █

blank Display data set list      P Print data set list
  V Display VTOC information      PV Print VTOC information

Enter one or both of the parameters below:
Dsname Level . . . SMPE.ZOS31
Volume serial . . .
  
```

```

DSLIST - Data Sets Matching SMPE.ZOS31      Row 1 of 11
Command ==> █      Scroll ==> PAGE

Command - Enter "/" to select action      Message      Volume
-----
SMPE.ZOS31.CSI                          *VSAM*
SMPE.ZOS31.HOLDDATA                     T31VS1
SMPE.ZOS31.IDX.INDEX                    T31VS1
SMPE.ZOS31.KSD.DATA                     T31VS1
SMPE.ZOS31.SMPLOG                       T31VS1
SMPE.ZOS31.SMPLOGA                      T31VS1
SMPE.ZOS31.SMPLTS                       T31VS1
SMPE.ZOS31.SMPMTS                       T31VS1
SMPE.ZOS31.SMPPTS                       D31VS1
SMPE.ZOS31.SMPSCDS                      T31VS1
SMPE.ZOS31.SMPSTS                      T31VS1
***** End of Data Set list *****
  
```

We mentioned that many SYSMODS are OCO – object code only – but it's also possible to install and build source. In those cases, data set SMPMTS contains macros used in assembling the source code, and SMPSTS contains source code .

DATA SETS ASSOCIATED WITH SMPE.ZOS31

```

Data Set List Utility

Option ==> █

blank Display data set list      P Print data set list
  V Display VTOC information      PV Print VTOC information

Enter one or both of the parameters below:
Dsname Level . . . SMPE.ZOS31
Volume serial . . .
  
```

```

DSLIST - Data Sets Matching SMPE.ZOS31      Row 1 of 11
Command ==> █      Scroll ==> PAGE

Command - Enter "/" to select action      Message      Volume
-----
SMPE.ZOS31.CSI                          *VSAM*
SMPE.ZOS31.HOLDDATA                      T31VS1
SMPE.ZOS31.IDX.INDEX                     T31VS1
SMPE.ZOS31.KSD.DATA                      T31VS1
SMPE.ZOS31.SMPLOG                        T31VS1
SMPE.ZOS31.SMPLOGA                       T31VS1
SMPE.ZOS31.SMPLTS                        T31VS1
SMPE.ZOS31.SMPMTS                        T31VS1
SMPE.ZOS31.SMPPTS                        D31VS1
SMPE.ZOS31.SMPSCDS                       T31VS1
SMPE.ZOS31.SMPSTS                        T31VS1
***** End of Data Set list *****
  
```

You can tell SMP/E to hold a SYSMOD if it's not a good idea to install it immediately. There may be installation-specific dependencies that have to be dealt with before it's safe to apply the SYSMOD.

When that happens, SMP/E saves the Reason Report from APPLY processing in the HOLDDATA data set.

HOLDDATA EXAMPLE

```
BROWSE      SMPE.ZOS31.HOLDDATA(TGT31A) - 01.00      Line 0000000000 Col 001 080
Command ==> |                                     Scroll ==> CSR
***** Top of Data *****
BYPASSED HOLD REASON REPORT FOR APPLY PROCESSING

TYPE      REASON ID  FMID      SYSMOD    ++HOLD DATA
-----
SYSTEM    ACTION    HBB77E0  UJ93618  ++HOLD(UJ93618) SYSTEM FMID(HBB77E0) REASO
                                COMMENTC
```

```
++HOLD DATA
-----
++HOLD(UJ93618) SYSTEM FMID(HBB77E0) REASON(ACTION) DATE(23205)
COMMENTC
*****
* FUNCTION AFFECTED: BCP    SMF                      (0A65166) *
*****
* DESCRIPTION      : Recompile/relink application          *
*****
* TIMING           : Post-Apply                            *
*****
With this PTF, the IFAQUERY executable macro code is updated.
Programs that call IFAQUERY must be recompiled to include the
updated macro code provided by this PTF. A program that is
compiled with the updated macro code can run compatibly on a
system that does not have the PTF for 0A65166 applied.
).

++HOLD(UJ93702) SYSTEM FMID(HBB77E0) REASON(ACTION) DATE(23220)
COMMENTC
The Stand-Alone Dump IPL volume must be rebuilt.
).
```

Here's a snippet from member TGT31A in a HOLDDATA data set. There's a lot more content that looks like this.

The lower screenshot is the same member after scrolling to the right so we can see all the text of the report.

The report cites a predefined reason category, in this case "action," identifies the system components affected, and provides an explanation of why the SYSMOD is held.

For reason "action," the report describes the actions necessary before the APPLY can be completed, or after the APPLY but still during the installation process.

RUNNING SMP/E IN BATCH

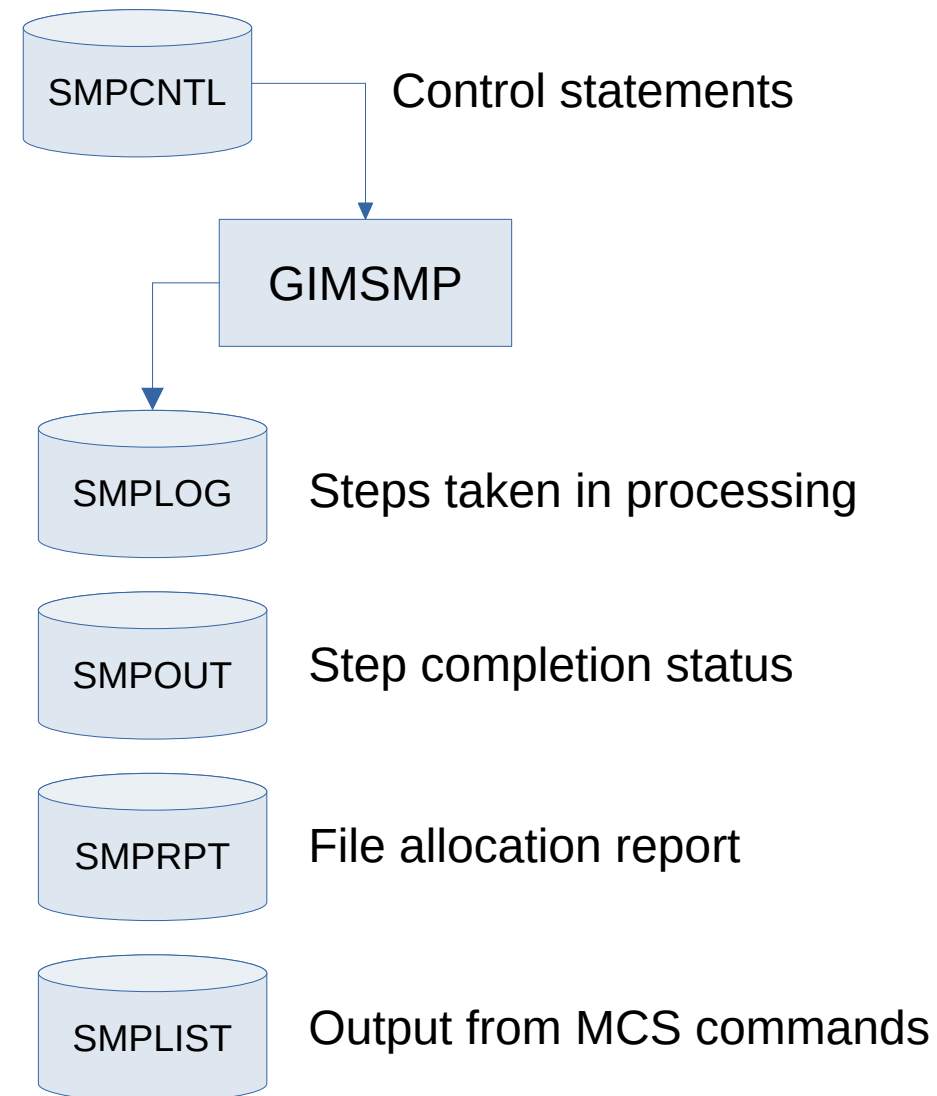
SMP/E is a batch utility program. When people talk about SMP/E "commands" they're referring to the source statements in a declarative language called MCS (Modification Control Statements), and not to individual commands entered interactively.

The name of the program is GIMSMP – GIM is the IBM product prefix for SMP/E. GIMSMP reads the control statements (or "commands") from a data set with DD name SMPCNTL.

SMPCNTL contains 80-byte card-image records. As with control statements for other IBM utilities, you can code the commands anywhere between columns 1 and 72.

```
//SMPCNTL DD *
SET BOUNDARY(DLB31).
LIST CLIST(ACBQADG4).
/*
```

DDNAME	StepName	ProcStep	DSID	Owner
JESMSG LG	JES2		2	IBMUSER
JESJCL	JES2		3	IBMUSER
JESYSMSG	JES2		4	IBMUSER
SMPLG	VERIFY		102	IBMUSER
SMPOUT	VERIFY		103	IBMUSER
SMRPT	VERIFY		104	IBMUSER
SMPLIST	VERIFY		105	IBMUSER



RUNNING SMP/E IN BATCH

```
CSI QUERY - CLIST      ENTRY      Row 1 to 1 of 1
SCROLL ==> CSR

To return to previous panel, enter END .

Primary Command: FIND

Entry Type:  CLIST      Zone
Entry Name:  ACBQADG4   Zone

                                LASTUPD: HDZ3310

FMID      HDZ3310
RMID      HDZ3310
DISTLIB   ADGTCLIB
SYSLIB    DGTCLIB
-----
***** Bottom of data *****
```

SMP/E ISPF query output

Same query in batch,
contents of SMPLIST.

```
SDSF OUTPUT DISPLAY IBMUSERZ JOB01456  DSID   105 LINE 0      COLUMNS 02- 81
COMMAND INPUT ==>
***** TOP OF DATA *****
PAGE 0001  - NOW SET TO DLIB   ZONE DLB31   DATE 06/24/26   TIME 06:10:05   SMP/E

LIST CLIST(ACBQADG4).
PAGE 0002  - NOW SET TO DLIB   ZONE DLB31   DATE 06/24/26   TIME 06:10:05   SMP/E

DLB31  CLIST  ENTRIES

NAME

ACBQADG4  LASTUPD      = HDZ3310  TYPE=ADD
          LIBRARIES    = DISTLIB=ADGTCLIB  SYSLIB=DGTCLIB
          FMID         = HDZ3310
          RMID         = HDZ3310

PAGE 0003  - NOW SET TO DLIB   ZONE DLB31   DATE 06/24/26   TIME 06:10:05   SMP/E

LIST      SUMMARY REPORT FOR DLB31

ENTRY-TYPE  ENTRY-NAME      STATUS

CLIST      ACBQADG4          STARTS ON PAGE 0002
***** BOTTOM OF DATA *****
```

RUNNING SMP/E IN BATCH

This is the JCL that produced the output shown on the preceding slides.

DDNAME SMPCSI identifies the CSI we're operating against.

You can see the DD statements for the output data sets we saw in the SDSF screenshot.

The MCS narrows the query down to zone DLB31, Entry Type "CLIST", and Entry "ACBQADG4."

```

EDIT          IBMUSER.LAB.JCL(GIMZOS31) - 01.03          Columns 00001 00072
Command ==>                                           Scroll ==> CSR
***** Top of Data *****
000001 //IBMUSERZ JOB , 'LIST TARGET ZONE',
000002 //          CLASS=A,MSGCLASS=X,MSGLEVEL=(1,1),REGION=0M,
000003 //          NOTIFY=&SYSUID
000004 //VERIFY EXEC PGM=GIMSMP
000005 //SMPCSI DD DSN=SMPE.ZOS31.CSI,DISP=SHR
000006 //SMPLOG DD SYSOUT=*
000007 //SMPDUT DD SYSOUT=*
000008 //SMPRPT DD SYSOUT=*
000009 //SMPLIST DD SYSOUT=*
000010 //SMPPTS DD DSN=SMPE.ZOS31.SMPPTS,DISP=SHR
000011 //SYSPRINT DD SYSOUT=*
000012 //SMPCTL DD *
000013 SET BOUNDARY(DLB31).
000014 LIST CLIST(ACBQADG4).
000015 /*
***** Bottom of Data *****

```

For queries in batch, you want to narrow the search as much as you can, because the job may run a long time and generate huge amounts of spool data if the query is too broad.

Rule of thumb: Use ISPF for queries when you aren't sure what's in a CSI. Run GIMSMP in batch when you're ready to apply system modifications.

THE SYS1.SAMPLIB DATA SET

```
VIEW      SYS1.SAMPLIB
Command ==> 
```

Name	Prompt	Size	Created
. GIMALLC			
. GIMASAMP			
. GIMCRSAM			
. GIMCSAMP			
. GIMDDALC			
. GIMDDDEF			
. GIMEXITS			
. GIMISMKD			
. GIMMKDIR			
. GIMOPCDE			
. GIMPRSAM			
. GIMPSAMP			
. GIMSAMPU			

Many of the products IBM and others ship for the z/OS environment are pretty complicated to install, configure, and use. SMP/E itself is like that, and so are many of the IBM products and updates we might need to install.

Fortunately, we have the SYS1.SAMPLIB data set to help us.

Most of the components that ship with z/OS and most of the products we might install provide members for the SYS1.SAMPLIB library.

These contain documentation, installation guides, sample JCL snippets, sample PARMLIB members or snippets, and other useful things.

A day in the life of a z/OS system programmer often includes building jobs based on members in the sample library. The sample members are usually not runnable without some tweaking, and many of them contain only bits and pieces of jobstreams, configuration settings, and other items. They also contain copious comments, instructions, and information about dependencies and gotchas.

The product prefix for SMP/E is GIM. The screenshot shows members in SYS1.SAMPLIB whose names begin with GIM. They pertain to SMP/E in one way or another.

SMP/E ELEMENT TYPES

Program Object	PROGRAM
Load Module	MOD
Source (Assembler, COBOL, JCL, etc.)	SRC
Macro	MAC
JCL Procedure	PROC
CLIST	CLIST
REXX exec	EXEC
ISPF panel	PNL
ISPF skeleton	SKL
ISPF message member	MSG
ISPF help/tutorial text	HLP
General text member	TEXT

WHAT YOU KNOW AND WHAT YOU CAN DO

- You know the role of SMP/E in z/OS system management.
- You can locate and browse information in SMP/E data sets using ISPF panels.
- You are somewhat familiar with the kinds of information stored in each SMP/E data set and how SMP/E uses the information.

WHAT'S NEXT

- z/OSMF – z/OS Management Facility